Global Disaster Data Analysis

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***Abstract*—Using information from the World Development Indicators (WDI), (EMDAT), (UNDP) for the years 1975–2025, this study offers an exploratory data analysis of disasters worldwide. For a total of 187 countries, the analysis looks at the connections between disaster impact index, vulnerability, resilience recovery, GDP, aid flow among other variables. We find important trends using thorough data pre-processing and visualization methods, such as differences in regions, countries, social development indicators, the amount of aid flow received and recovery. The results show that developing countries are disproportionately affected by disasters and recover less quickly as opposed to developed countries which are more resilient. Developing countries receive more aid flow but despite this are not able to recover.**

***Index Terms*—Social Development Index, Natural Disasters, Worldwide Analysis**

I. Introduction

Disasters impact humanity across multiple dimensions every year. These include loss such as injuries and deaths, economic loss and social disruption. Traditionally, responses to disasters focus on immediate rather than systemic resilience. A quantitative framework is needed to indentify vulnerabilities across regions,to create strategic plans to handle national and international level disasters in the future. This can be done by analyzing past disasters, socio economic standing, human and economic loss and governance indicators.

1) Background: The increase in the frequency and severity of disasters as result of Climate Change, Population Growth, and Urbanization (COPG) For the past five decades, Global Economic Losses have risen sharply, with developed countries generally losing more than developing nations. Losses occur across geographic boundaries due to a Global Supply Chain System and Urban areas that are densely populated. When a disaster strikes, the impact can be felt beyond the boundaries where it occurred; therefore, understanding both immediate and long-term impacts of a disaster on an area is necessary for creating an effective Recovery Plan. Data driven approaches provide a method for comparison of Regions and enable identification of Systemic (Institutional) Weaknesses that would be otherwise ignored in conventional reporting methods.

2) Problem Statement: Global disasters and their impact are on a steady rise. From wildfires to floods to earthquakes, vulnerabilities in human systems expose millions across the world to danger. This project designs an analytical dashboard to quantify and visualize disasters across the world in the last 5 decades. The system will help understand the socio-economic, governance and climate factors that drive resilience, vulnerability and disaster impact.

3) Objectives: The objectives of this visualization project are the following.

1) Integrate data from multiple open-source datasets to analyze trends across multiple variables.
2) Define and model Resilience, Vulnerability, Disaster Impact Index (DII), Resilience Recovery Score (RRS), and Composite Resilience Index (CRI).
3) Design and implement a dynamic Tableau dashboard that is interactive and tells a story.
4) Filter graphs on Country Name, Years, Region, Socio-economic Index and Disaster Type.
5) Support decision-making and policy-making through actionable visual insights translated from the dashboard.

II. Data Sources and Data Preprocessing

A. Data Sources

Data were obtained from the following sources. The original source is referenced in the appendix. We acknowledge the contribution of the following open-source datasets in our project.

1) EM-DAT: Disaster frequency, fatalities, affected population, and economic damage [2].
2) World Bank: GDP, GDP growth, socio-economic indicators, CO_2 emissions [1].
3) UNDP: Human Development Index (HDI)[3].

B. Data Pre-processing

1) Data Challenges: A total of 7 datasets were used to make a master dataset. Seperate data sets for types of disasters, number of people left homeless from disasters, number of deaths, total economic damages, total people affected and injured by disasteres were 5 seperate datasets taken from EMDAT. Human Development Index (HDI) statistics up to the year 2025 were taken from UNDP. Finally, socio-economic indexes, GDP, access to electricity, literacy rate, carbon dioxide emissions, aid flow, urban population percentage, forest area, political stabilty and absense of terrorism were obtained from World Band Data. The collection of three separate international datasets increased the complexity of the different structures and timeframes used by each source for recording their information. For example, every country reports its information in differing formats from previous years, many also only provide partial values when reporting like dropping some years from their submission. Additionally, some of the information was related to past events, so the ability for each information provider to collect, verify, validate, and report that information was dependent upon their capacity within their local jurisdiction. Also, indicators of governance data were typically updated less frequently than the data related to economics and demographics. Thus, this required careful attention to ensure alignment. Finally, addressing all of these differences was necessary when constructing composite indices that are used later within the analysis process.

2) Merging Data Sources: The data pre-processing pipeline converted data from multiple sources into a single data frame. To make UNDP and World Bank Data merge and be consistent with EMDAT data each dataset was reshaped and melted to provide a unified structure. This step was critical for building time series models and computing lagged variables later down the line. The variable Year was converted to `int` data type to prevent merge failures and ensure consistent sorting and grouping. The data frame was merged on the keys Country and Year. This merge was crucial to enrich the data frame with HDI, GDP, disaster metrics and socio-economic indicators, among others. This data frame was then further preprocessed, cleaned, explored and feature engineered.

3) Exploratory Data Analysis: First the data were merged based on the composite index Time, Country. The total rows were 22,140 and the total columns were 26. Missing values, duplicate rows, and summary statistics were explored. Fourteen columns were string while fifteen were numerical.

4) Clean Country Names: Country Name column was preprocessed by using a list of valid countries and a dictionary for remapping countries from their old names to their current names. Territories, unclassified regions and disputed countries were removed. After cleaning, 187 countries were left.

5) Data Types and Duplicates: Data types were corrected to numerical and categorical as required. At the end 19 columns were numerical and 6—Country Name, Country Code, Time Code, Region, Disaster Type, and Socio-Economic Group—were categorical.

All duplicated rows were dropped and asserted.

6) Missing Values Treatment: Missing values existed due to missing years, incomplete Human Development Index records and countries without disasters in a given year, among other factors. Multiple approaches were used to handle missing values.

Missing values were handled by replacing special characters such as ".." with `NaN`. For numerical columns, missing values were replaced with mean. For categorical columns, mode was used. For disaster metrics, zero-imputation was used for when no disasters took place in a country. Literacy rate was dropped entirely as it had more than 50% null values.

Socio-economic group column was filled in using country-wise mode. Years before 1975 were removed. Region was filled in based on the Country Name which was mapped through a custom dictionary.

Columns that do not change drastically over a short period such as access to electricity, carbon dioxide emissions, forest area, GINI index, urban population, inequality index and Human Development Index records were forward filled.

7) Feature Engineering: Multiple new columns were made to improve analysis performance and to model disaster impact and response.

1) **Disaster Exposure** (E): Frequency and severity of disasters in a region.
2) **Vulnerability** (V): Human and economic losses relative to population or GDP.
3) **Adaptive Capacity** (A): Socio-economic and institutional strength for recovery.
4) **Disaster Severity Weight** based on the type of disaster.
5) **Disaster Impact Index (DII)**:
$$\text{DII} = \frac{\text{Total Deaths} + \text{People Affected}}{\text{GDP per Capita}} \times \text{Disaster Severity Index}$$
6) **Time to Recovery**:
$$T_{\text{recovery}} = \text{Years required to reach pre-disaster output}$$
7) **Resilience Recovery Score (RRS)**:
$$\text{RRS} = \frac{(\text{GDP Growth}_{\text{post}} - \text{GDP Growth}_{\text{pre}})}{T_{\text{recovery}}} + \frac{\text{HDI} + \text{Gov Index}}{2}$$
8) **Composite Resilience Index (CRI)**:
$$\text{CRI} = \frac{A}{E} \times V$$

III. Methodology

A. Tools used

Python was used to preprocess the dataframes. The libraries used were pandas and skimpy. Pandas was used to make the data frame and skimpy was used to organize and beautify the statistical output that describes the data frame.

The tool used to visualize all graphs is Tableau Desktop Public. The following graphs were made using it's built in features:

1) Visualization Techniques:

1) **Bubble Chart**: Illustrates the corelation between wealth operationalized through Average GDP and Average Resilience Recovery Score for each Nation. The bubble size depicts Disaster Impact Index (DII)
2) **Choropleth Map**: Displays global resilience geographically across time
3) **Stacked Bar Chart**: Compares the Resilience Recovery Score for top 10 performing nations across disaster types.
4) **Stacked Area Chart**: Visualizes bilateral aid flow over two decades for the top 10 most vulnerable countries.
5) **Scatter Plot**: Visualizes the corelation between vulnerability and exposure using logarithmic axes.
6) **Timeseries Trellis Plot**: Displays each region's Resilience Recovery Score across time.
7) **Treemap**: visualize large amounts of hierarchical data and to show the part-to-whole relationship between Bilateral Aid flow, Disaster Types and Damage.

Filters across disaster type, time, region, country and socio-economic group were applied to all graphs. Graphs were animated by years.

These visulizations were picked to show both micro and macro trends across various attributes. Through the use of bubble charts, comparison of countries across the world can be achieved while simultaneously controlling for other variables through the size of the bubbles. The clarity of spatial trends presented in choropleth maps is superior for visualizing geographic data. As disaster-related data can often be extreme outliers, logarithmic scatter plots were chosen to create this visual. The design decisions of the bubble pattern and logarithmic scatter were designed to maintain interpretability on the trends while allowing for continued variability. A dashboard allows the user to view and analyze regional and global patterns while providing context to the individual countries.

IV. Results

A. Key Results

1) Risk and Vulnerability Assessment: The scatter plot shows that high disaster combined with low vulnerability is rare. There exists a positive corelation between the two variables across time. Japan, however, is an exception. It exhibits high economic and institutional capacity to reduce vulnerability, despite facing severe disaster exposures like Tsunamis and Storms.

2) Disproportionate Resilience Based on Socio-economic Index: The choropleth map shows that countries that are developed are likely to be more resilient to national disasters and damage. This is as opposed to countries that are still developing and are economically struggling. This phenomena means that developing nations struggle to allocate resources, have weaker social and infrastructural capacities to deal with national disasters. This increases their vulnerability. The top 10 vulnerable countries are also all developing countries, other than China, as shown in the area chart.

3) Drivers of Resilience and Recovery: To answer if wealthier nations recover faster or if governance matters more, our analysis found that the answer is nuanced. Both matter for various reasons. Wealth of GDP establishes a baseline. Nearly all wealthy nations have a positive resilience recovery score (RRS) confirming that high income provides a strong foundation for a resilient nation.

Among low income countries, there is a lot of variance due to other factors like institutional stability, HDI, Gini Index and strong governance.

The most resilient nations have a strong balance of all the above strenghts. The key is not a single factor but a multitude of factors that increase resilience to disasters.

Aid flow has also showed an upward trend for the top 10 major recipient countries from 2003. However, this aid has plateaued after 2015. This suggests that the total volume of aid is not keeping pace with the increasing severity and frequency of disasters. Total Aid flow on a treemap diagram shows that countries that receive the most aid are not the ones with the most total damage. Developed countries like China face severe disaster impact but do not receive the same intensity of bilateral aid flow.

4) Cross-Regional Variability: A trend across all graphs is the gap between regions. The differences between the regions of Asia and Africa in terms of their higher exposure combined with recovery rates, versus Europe/North America's sustainability of resilience scores, continue to exist over many decades. This is despite developing nations experiencing significant economic growth. This suggests that there is a need for further investigation of causes which go beyond income.

B. Visualization and Interpretation

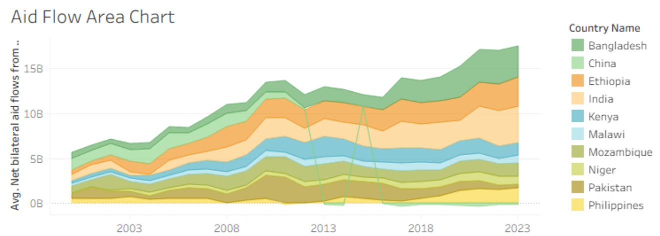

Fig. 1. Aid Flow Area Chart: Average Net Bilateral Aid Flow Area Graph for Top 10 Vulnerable Nations

The Visualization in figure 1 revealed that over a long period of time across the top 10 most vulnerable nations Bilateral Aid flow has increased. The only major outlier in the pattern is China, which seems a dip twice between the years of 2012-2013 and 2016-2017.

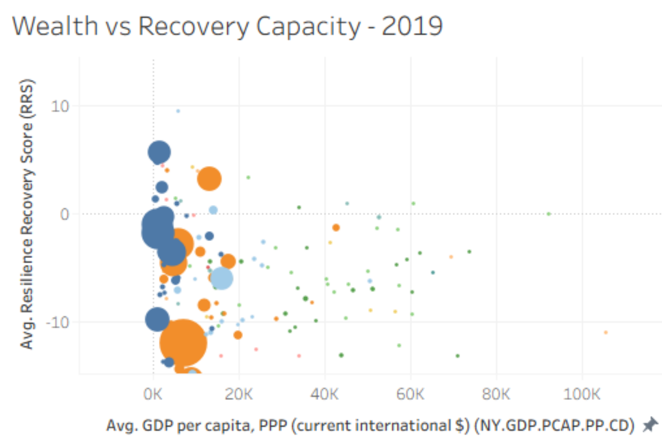

Fig. 2. Wealth vs Recovery Capacity: a Bubble Chart of Average GDP per Capita and Average Resilience Recovery Score across Time

The wide vertical spread for poorer countries is the most intriguing finding shown in figure 2. It hows that effective policies and smart resource allocation can help some low-income nations be resilient to disasters and overcome national economical issues.

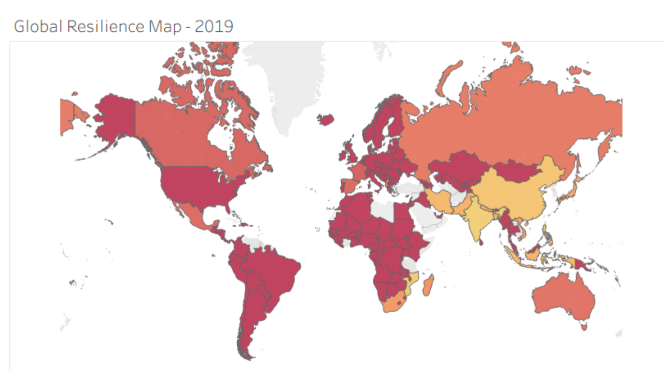

Fig. 3. Global Resilience Map: Choropleth Map for Resilience Across Nations with Time

The geographical map in figure 3 paints a clear picture. There is a strong corelation between economic development, strong governance and resilience of a nation. Developing nations show low resilience and require long-term investments in infrastructure and social systems to increase resilience.

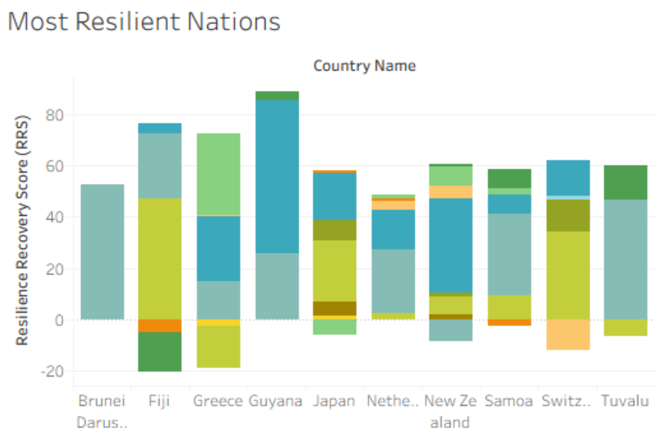

Fig. 4. Most Resilient Nations: A stacked Bar Chart of the top 10 resilient nations across Disaster Types with Resilience Recovery Scores

The Stacked Bar Chart graph in figure 4 breakdowns the top 10 resilient nations. It analyzes the multi-factorial reasons for the nations resilience. Analyzing the top nations; Fiji, Greece, Netherlands etc provides a way to identify economic and social strengths that increase resilience.

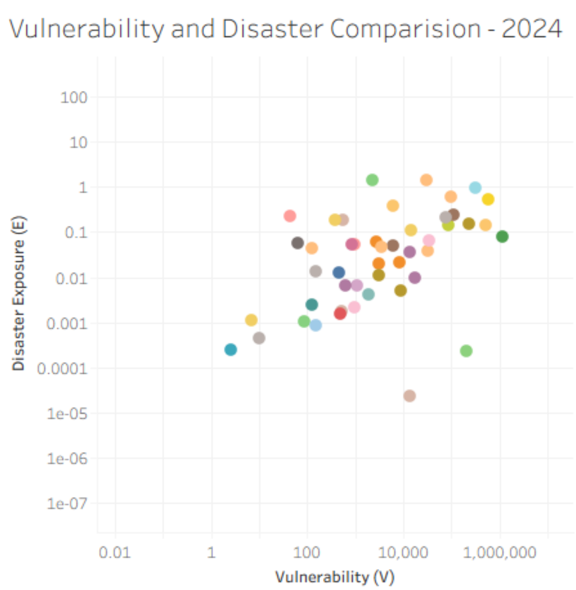

Fig. 5. Vulnerability and Disaster Comparison: A logarithmic scatter plot to visualize countries with low Vulnerability and high Disaster Exposure

In figure 5, the use of logarithmic scale emphasizes that the differences in vulnerability are vast. High exposure does correlate with high vulnerability. Japan, however, is a country with low vulnerability and high disaster exposure. The scatterplot helps find the corelation between two numerical attributes; Vulnerability (V) and Disaster (D).

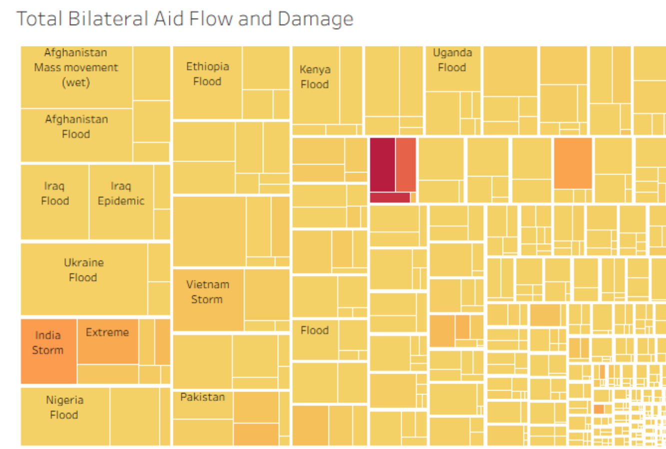

Fig. 6. Aid Flow and Damage Treemap: size represents Average Net Bilateral Aid Flow and intensity on a gradient represents Total Damage in US$

Figure 6 shows how disaster impact is disproportionate. The high damage concentration blocks are few and need critical, large-scale focus. However, they have low aid flow. Each treemap block is further divided into blocks based on disaster type. Countries typically have more than one disaster type occur in their boundaries. This means that the human loss and infrastructure damage varies greatly. Hence, the aid needed is also specific. The graph shows that across time Afghanistan has received the most Aid. Due to war and the geography of the region, Afghanistan has been hit with floods, storms, extreme temperatures, earthquakes and has seen mass movement. On the other hand, countries like China have been more severely impacted by storms, earthquakes and extreme temperatures. However, they have not received the same influx of Aid flow.

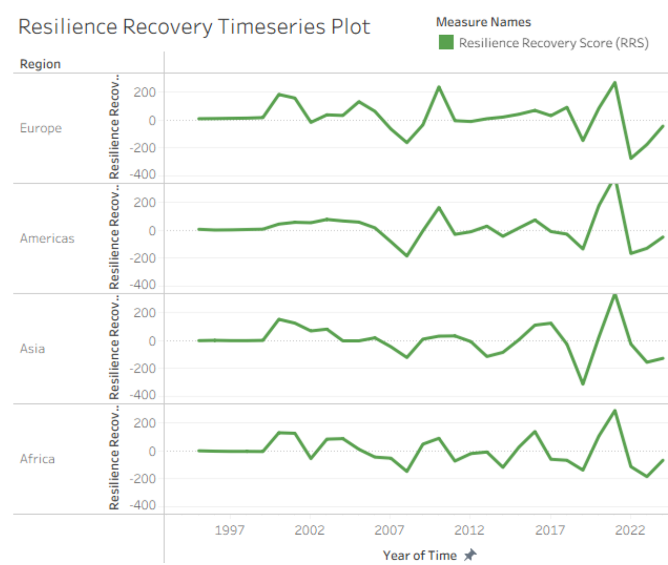

Fig. 7. Resilience Recovery Timeseries Plot: A region wise comparison of Resilience Recovery Index and across Time

Figure 7 shows the disproportionate difference in regions in Resilience Recovery Scores over time. While Africa remains lower Resilience Recovery Score values, regions like Europe and Americas are more resilient over time. In this analysis, the overall patterns establish that disaster resilience has a multidimensional nature. Economic capacity is undoubtedly an important factor in recovery; however, institutional strength and social development are equally important. In fact, countries' levels of GDP are not the sole determinants of disaster outcomes, proving that there are various other factors that affect disaster resilience. Finally, the increasing occurrence of more severe disasters indicates that existing historical patterns are likely not sufficient to predict future risks, demonstrating a need for improved modelling processes of risk assessment.

V. Limitations

There are multiple limitations of this project.

1) There are many data inconsistencies that limit the scope of the study. The inconsistencies are to be expected as they are the result of the data coming from heterogeneous measurement methodologies. Reporting bias may exist in the data and the missing values imputed may have simplified the results.
2) Model Indices such as DII, RRS and CRI do not fully capture the complexity of the problem. They do not take into account may environmental and economic interdependencies. Further variables could have been added to provide a more robust calculation of these indices.
3) Annual metrics are analyzed in this project. So the project fails to capture the immediate post-disaster dynamics. The indicators are lagging as the data is annual, which means cruical details needed for deeper analysis are missing.
4) Aid data is strictly bilateral. This excludes multilateral and in-kind contributions. These metrics were not considered but could have contributed to the project.
5) Missing disaster severity scale in the dataset. As this attribute is modeled in pre-processing and could not be found in the original dataset, this limits the validity of the findings.

VI. Recommendations and Future Work

A. Policy Recommendations

In light of these results, we advise:

- **Strengthen Early Warning Systems and Hazard Monitoring**: Countries with high Exposure and frequent disasters require strong forecasting. There need to be policies to expand real-time monitoring of floods, heatwaves,cyclones and droughts. There need to be mobile-based alert systems based on geolocation and increased coverage of communication in rural and low-literacy areas. These actions directly reduce fatalities and injuries by decreasing vulnerability (V).
- **Invest in Critical Infrastructure and Local Protective Measures**: Areas with high economical and physical damage need resilient infrastructure that is suitable for flood defenses, heatwaves and wind. This means strengthened transport systems, heat-resistant construction material, power grids, sea walls and drainage systems.
- **Target Social Protection Programs to Vulnerable Populations**: Human impact is tied to vulnerability. Therefore, governments should increase cash-transfer programs after disasters, expand health access to hazard prone areas, create shelters and prioritize disaster responses for disabled people.
- **Strenthen Governance and Institutional Capacity**: Government indicators shape disaster resilience. Hence, governments should improve transparency in disaster relief spending, create independent disaster risk assessment bodies and strengthen federal,provincial and local authorities. Governments should also focus on economic growth as metrics like GDP affect the resilience of a nation and its response to disaster management. Richer countries deal with disasters quicker and recover faster.

B. Future Research Directions

- To determine the causal relationships between model indices and metrics.
- Examine sub-national data to find differences within a country.
- Incorporate satellite imagery, climate forecasts and other dynamic data for real-time integration.
- Analyze the success of various policy initiatives in various nations to increase disaster resilience, such as China.

In nations across the globe, addressing disasters is essential for social stability, economic growth, and innovation. Although more research is required to develop and assess particular interventions, this analysis offers a basis for the creation of evidence-based policies.

VII. Conclusion

This project shows how combining international datasets can provide significant information regarding disaster vulnerability, exposure, and recovery. These findings indicate that there are geographic disparities in levels of disaster risk, and thus, socio-economic and institutional determinants are highly relevant for determining the ability of communities to be resilient to these risks. While there are limitations associated with the data, the conclusions provide a basis for developing policies aimed at building adaptive capacity, enhancing governance, and prioritizing long-term disaster risk reduction activities. Further research to gather more detailed and real-time data will be necessary in order to create better and more specific strategies for building resilience.

References

[1] World Development Indicators (WDI), 2024. World Bank DataBank. Available at: https://databank.worldbank.org/source/world-development-indicators
[2] EM-DAT, "The International Disaster Database." Available: https://www.emdat.be/
[3] UNDP, "Human Development Reports Data Center." Available: https://hdr.undp.org/data-center

Appendix A
Preprocessing Code

A. Acquiring and Loading Data

```
# Data manipulation
import datetime
import numpy as np
import pandas as pd
import pandas.api.types as ptypes
from skimpy import skim, clean_columns

# Visualizations
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline

# Pandas settings
pd.options.display.max_columns = None
pd.options.display.max_colwidth = 60
pd.options.display.float_format = '{:,.3f}'.format

# Visualization settings
from matplotlib import rcParams
plt.style.use('fivethirtyeight')
rcParams['figure.figsize'] = (16, 5)
rcParams['axes.spines.right'] = False
rcParams['axes.spines.top'] = False
rcParams['font.size'] = 12
rcParams['savefig.dpi'] = 300
plt.rc('xtick', labelsize=11)
plt.rc('ytick', labelsize=11)
custom_palette = ['#003f5c', '#444e86', '#955196', '
    #dd5182', '#ff6e54', '#ffa600']
custom_hue = ['#004c6d', '#346888', '#5886a5', '#7
    aa6c2', '#9dc6e0', '#c1e7ff']
custom_divergent = ['#00876c', '#6aaa96', '#aecdc2',
     '#f1f1f1', '#f0b8b8', '#e67f83', '#d43d51']
sns.set_palette(custom_palette)
%config InlineBackend.figure_format = 'retina'

# Load DataFrame
df = pd.read_csv('merged_data.csv')

# Show rows and columns count
print(f"Rows␣count:␣{df.shape[0]}\nColumns␣count:␣{
    df.shape[1]}")

# Show data types
df.info()

# Missing values
missing_percent = df.isna().mean().sort_values(
    ascending=False)
print('----␣Percentage␣of␣Missing␣Values␣(%)␣-----')
if missing_percent.sum():
    print(missing_percent[missing_percent > 0] *
        100)
else:
    print('None')

# Check for duplicates
df_dedup = df.drop_duplicates(keep=False)
print(f"No.␣of␣entirely␣duplicated␣rows:␣{len(df)␣-␣
    len(df_dedup)}")
df[df.duplicated(keep=False)].sort_values(['Country␣
    Name', 'Time']).head(10)

# Drop duplicates by key columns
df_dedup = df.drop_duplicates(subset=['Country␣Name'
    , 'Time', 'Disaster␣Type'], keep=False)
print(f"No.␣of␣entirely␣duplicated␣rows:␣{len(df)␣-␣
    len(df_dedup)}")

# Clean columns
df = clean_columns(df)
```

```
# Rename columns
df = df.rename(columns={
    "gdp_per_capita_ppp_current_international_": "
        gdp_per_capita_ppp",
    "access_to_electricity_of_population
␣␣␣␣_eg_elc_accs_zs": "access_to_electricity",
    "literacy_rate_adult_total_
␣␣␣␣of_people_ages_15_and_above_se_adt_litr_zs": "
    literacy_rate_adult_total",
    "carbon_dioxide_co2_emissions_excluding
␣␣␣␣_lulucf_per_capita_t_co2e_capita_
␣␣␣␣en_ghg_co2_pc_ce_ar5": "
    carbon_dioxide_co2_emissions",
    "net_bilateral_aid_flows_from_dac_
␣␣␣␣donors_total_current_us_dc_dac_totl_cd": "
    net_bilateral_aid_flows",
    "urban_population_of_total_population
␣␣␣␣_sp_urb_totl_in_zs": "urban_population_percent",
    "forest_area_of_land_area_ag_lnd_frst_zs": "
        forest_area_percent",
    "political_stability_and_absence_of_
␣␣␣␣violence_terrorism_estimate_pv_est":
        "political_stability_and_absence_of_
␣␣␣␣␣␣␣␣violence_terrorism_estimate",
    "total_deaths_emdat": "total_deaths",
    "no_injured_emdat": "no_injured",
    "no_affected_emdat": "no_affected",
    "no_homeless_emdat": "no_homeless",
    "total_affected_emdat": "total_affected",
    "total_damage_current_us_emdat": "total_damage",
    "total_damage_adjusted_current_us_emdat": "
        total_damage_adjusted",
    "dis_no_emdat": "disaster_count",
    "gini_index_si_pov_gini": "gini_index",
    "time": "Time",
})
```

B. Cleaning and Harmonizing Country Names

```
print("Before␣cleaning:", df["country_name"].nunique
    ())

def clean_country(country):
    if any(item in str(country) for item in
        remove_if_contains):
        return np.nan
    if country in rename_map:
        return rename_map[country]
    return country

df["country_name"] = df["country_name"].apply(
    clean_country)
df = df[df["country_name"].notna()]

print("After␣cleaning:", df["country_name"].nunique
    ())
```

C. Changing Data Types and Removing Duplicates

```
numeric_cols = df.select_dtypes(include="number").
    columns.difference(["time"])
for col in numeric_cols:
    df[col] = pd.to_numeric(df[col], errors='coerce'
        )

df = df.drop_duplicates(keep='first')
```

D. Filling Missing Socioeconomic Groups

```
developing_countries = ['Eswatini', 'Micronesia', '
    North Korea', 'Tanzania']
developed_countries = ['Monaco']

def fill_socio_economic(group):
    mode_value = group['socio_economic_group'].mode
        ()
    if not mode_value.empty:
        group['socio_economic_group'] = group['
            socio_economic_group'].fillna(mode_value
            [0])
    else:
        country_name = group.name
        if country_name in developing_countries:
            group['socio_economic_group'] = group['
                socio_economic_group'].fillna('
                Developing')
        elif country_name in developed_countries:
            group['socio_economic_group'] = group['
                socio_economic_group'].fillna('
                Developed')
    return group

df = df.groupby('country_name').apply(
    fill_socio_economic)
```

E. Filling Missing Regions

```
df['region'] = df['region'].fillna(df['country_name'
    ].map(country_to_region))
```

F. Filling Missing Disaster Types

```
disaster_cols = [
    "total_deaths", "no_injured", "no_affected", "
        no_homeless",
    "total_affected", "total_damage", "
        total_damage_adjusted",
]

df.loc[
    (df["disaster_type"].isna()) &
    (df[disaster_cols].sum(axis=1) == 0),
    "disaster_type"
] = "None"
```

G. Final Cleaning Steps

```
if "country_name" in df.index.names:
    df = df.reset_index(level="country_name", drop=
        True)

df = df.reset_index(drop=True)

slow_changing_cols = [
    "socio_economic_group", "gini_index", "
        inequality_index", "hdi",
    "access_to_electricity", "
        literacy_rate_adult_total",
    "forest_area_percent",
    "
        political_stability_and_absence_of_violence_terrorism_estimate
        ",
]

df[slow_changing_cols] = (
    df.groupby("country_name")[slow_changing_cols]
    .transform(lambda g: g.ffill().bfill())
)
```

```
df = df.sort_values(["country_name", "time"])

missing_percent = df.isna().mean().sort_values(
    ascending=False)
print('---- Percentage of Missing Values (%) -----')
print(missing_percent[missing_percent > 0] * 100)
```

Appendix B
Datasets

Table I shows an excerpt of the original dataset after it was merged. The full CSV file is provided as supplementary material.

The table II shows a sample of the dataset after preprocessing. The full XLSX format file is provided as supplementary material.

TABLE I
EXCERPT OF RAW DATASET

Country Name	Country Code	Time	Time Code	GDP per capita PPP	Access to electricity	Literacy rate	CO2 per capita	Net aid flows	Urban population
Afghanistan	AFG	2000	YR2000	813.550255989624	4.4	..	0.0504760801948225	105239999.111742	22.078
Afghanistan	AFG	2001	YR2001	747.688045492413	9.3	..	0.0465729492262171	368649994.671345	22.169
Afghanistan	AFG	2001	YR2001	747.688045492413	9.3	..	0.0465729492262171	368649994.671345	22.169
Afghanistan	AFG	2002	YR2002	926.507941011646	14.1	..	0.0440777829029563	1134419991.01639	22.261
Afghanistan	AFG	2002	YR2002	926.507941011646	14.1	..	0.0440777829029563	1134419991.01639	22.261
Afghanistan	AFG	2002	YR2002	926.507941011646	14.1	..	0.0440777829029563	1134419991.01639	22.261
Afghanistan	AFG	2002	YR2002	926.507941011646	14.1	..	0.0440777829029563	1134419991.01639	22.261
Afghanistan	AFG	2002	YR2002	926.507941011646	14.1	..	0.0440777829029563	1134419991.01639	22.261
Afghanistan	AFG	2002	YR2002	926.507941011646	14.1	..	0.0440777829029563	1134419991.01639	22.261

TABLE II
EXCERPT OF PROCESSED DATASET

Index	Country Name	Time	AccessElectricity	CO2pc	DisNo	ForestArea	GDPpcPPP	Gini	HDI	InequalityIndex	AidFlows	NoAffected	NoH
69	Albania	2000	99.4	1.046446017	0	28.07664234	3976.926083	31.7	0.789	29.4	228990004.3	0	0
70	Albania	2001	99.4	1.123890708	0	28.12324818	4451.000524	31.7	0.789	29.4	214349993.8	0	0
71	Albania	2002	99.4	1.313138928	3	28.16985401	4840.347882	31.7	0.789	29.4	207800003.3	192110	0
72	Albania	2003	99.4	1.385438161	0	28.21645985	5169.842511	31.7	0.789	29.4	264279996.8	0	0
73	Albania	2004	99.4	1.458238835	2	28.26306569	5595.55058	31.7	0.789	29.4	216400002.6	2500	0
74	Albania	2005	99.4	1.377558661	2	28.30967153	6014.326823	30.6	0.789	29.4	257929998.1	400500	0